

ActInf Livestream #034.0 ~ "The free energy principle: it's not about what i

Livestream | 1:25:17

[Music]

[Music]

all right hello and welcome everyone
it's active lab live stream number 34.0
on december 3rd 2021 and we're
discussing the paper the free energy
principle it's not about what it takes
it's about what took you there
welcome to the actin flap
we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us
at the links here on this slide
this is a recorded and an archived live
stream so please provide us with
feedback so that we can improve on our
work
all backgrounds and perspectives are
welcome here and we'll be following
cool diatic video chat etiquette today
at this coda site you can find our
sortable searchable interactable and
growing database of past and future live
streams
and if nothing else isn't this paper
going to be about the past and the
future
and it's also our last paper discussion
that we're kicking off for 2021 in
number 34. so it's a pretty cool
discussion and i'm really looking
forward to it
in actin stream number 34 we're going to
be discussing the paper the free energy
principle it's not about what it takes

it's about what took you there by axel

constant in 2021

and just like all the other dot zeros

the video is a introduction to some of
the ideas and the context for the paper

it's not a review or an evaluation or a
final word

we'll go over the aims and the claims
the abstract and the roadmap of the
paper and then there's a couple keywords
including some that we haven't seen
before

and then we'll walk through the key
example which is the numerical example
of a simple signal detecting bacterium
and we'll kind of talk through that
example and see where that takes us
and i'll just note as with all the other
dot zeros check out the paper if you
want to learn more and axel's writing is
really a pleasure to read and he has
deep insight into the fep

so we'll start with just

saying hello and

anything that we else wanted to say so

i'm daniel i'm a researcher in
california

and

what excited me about this paper and got
me excited to discuss it was i think
axel's provocative as well as grounded
and respectful

approach to dealing with what has and
hasn't been done

in the fep

and the example is so minimal and paired
down but also insightful that i think it
lends itself to a jumping off point for

many many other discussions
dean
yeah uh uh thanks daniel i'm dean i'm
i'm in calgary
and uh i think what excited me about
this was that i um i didn't start out
really understanding what the title of
the paper even meant
but then when there when there was a bit
of a
an inversion and a bit of a taking
things inside out and
i don't know if you want to describe
that as a reversal of what is typically
viewed as the
time continuum or not
um
it just opened up a whole bunch of uh
interesting ideas which i was
contemplating as a
as a program developer because
i knew that
um in order for
the idea of being able to go from an
inference i believed going from an
inference which is a very small time
frame through a prediction set and
finally into being able to get to a
place of modeling or understanding
co-variation as a as collective
uh this paper would probably help and
then with each section i went through
it was it was very helpful in terms of
being able to sort of bring this idea
just as an idea
not in terms of its practical
implications but just as an idea
to life so

[Music]

okay

i'm sure he'll rejoin

i'm gonna just read through some of the
big questions and

again these are just the ones that we
wrote down so all read the first two and
then maybe say something and then dean
you can take the second two so two of
the big questions that the paper raises
are

what is the free energy principle about
and what is the entailment relationship
between life

and the minimization of free energy or
another way to read that is what is the
entailment relationship between life and
models that emphasize the minimization
of free energy

and

i just love how the paper dances between
both of them

and figuring out what the fep exactly is
and isn't in its distinction from
general bayesian theories from active
inference

these are really crucial distinctions
dean

yeah so the the second two questions
were one one was from the author i'll
read the fourth one first from the
author are prior subjective or objective
under the fep

um i put the words

and with the question mark yes

because i actually have a question about
this he he comes to a conclusion but i'm
not

i'm not sure what the timeline is switch
leads back to bullet three which is
what can we say about relationship when
we shift from looking at what something
is

the comparison of life and free energy
principle to how long it lasts he talks
about duration or the when aspect of
that and that's the thing that kind of
adds a twist to this because you can
look at it as very momentary or you can
look at his things as very cyclic
obviously expression episodic and
diachronic and you've used kairos and
there's lots and lots of ways to bring
that in but when we do

[Music]

things change pretty quickly
and it's so funny you had to go to the
fourth one
but was it really the fourth question
if you read it third
right

there we go
so the paper was published on two two 2
2 1

could have waited a year and gone for
the total too but
i understand you got to get those papers
out

and a few of the key aims and claims so
first using the author's words the
entailment problem is the confusion in
the entailment relationship between free
energy minimization and life so if
you've never heard the word entailment
never heard free energy we're going to
go into some of these things but the

entailment problem is people being
confused about entailment that's a
problem in this paper

the author dissolves what's called the
entailment problem proposed here by
providing a numerical example of free
energy minimization in a hypothetical
organism

i conclude with some brief
epistemological knowledge oriented
remarks that may be of interest for
those who worry about the explanatory
power of the fep

and then this is the last words of the
paper which we'll return to

at the end of this live stream

the claim i set up at the start of the
paper is that free energy minimization
is not sufficient for life

mic drop

fidel true

to the unifying grip of hypotheses in
historical sciences the free energy
principle is meant to account for all of
that

you me and the organism under study in a
unifying fashion

any thoughts on those claims we'll be
exploring them and discussing them in
the

today yeah no the only thing that i
immediately went to is was it was a
question that i would ask people who
were sort of on the other side of
whatever it is that they were moving
towards in terms of a
sort of a professional
um

participation participation

and so the entailment thing for me was

is it opportunity

over method

meaning method entailing opportunity or

or is it

met

opportunity being over method or method

being entailed by opportunity

and that is the same kind of question

that that axel looks at here

so

depending on how you see that entailment

things go off in completely different

directions so let's go and get into the

paper now because yeah that's something

that you can't resolve until

post-diction

you can expect how it'll be resolved but

right

then you look back and it might have

been different so

i'll read the abstract and then feel

free to give a thought and of course

anyone watching along live is always

welcome to give comments and questions

in the live chat so the abstract is

i'll read um the first and the third one

philosophical writings on the free

energy principle in the life sciences

often give the impression that

minimizing free energy is sufficient for

life

but minimizing free energy is not a

sufficient condition for life

in fact one can perfectly well conceive

of a system that actively minimizes its

free energy and for this very reason

moves inexorably towards death

so

where does the assumption of this

entailment relation come from

there is indeed an entailment relation

but it goes the other way around life

entails minimizing free energy

but another way if you exist now under

the right conditions

it's because you've done something like

minimizing your free energy however

the question of whether you exist

tomorrow

cannot be settled purely by the

resorting to the fact that you'll

minimize your free energy in order to

get there

the simple point i make in this paper is

that the free energy principle is not

concerned with the sufficient conditions

of existence but rather what must have

been the case necessary what must have

been the case given that you exist it's

not about figuring out what it takes to

be alive it's about figuring out what

took you there

and i'd recommend people to put this in

a speed text-to-speech reader it reads

just so fantastically in a robotic voice

i rarely do that for papers

but i was on my i was on my exercise

bike listening to the robot talk about

this and it was just

it was hilarious

so here's the simple point in the paper

free energy principle is not concerned

with the sufficient conditions of

existence but rather with what must have

been the case given you exist it's worth
restating because it's a simple point
but the depth
of how far off one can be by missing
this point is vast
it's not about figuring out what it
takes to be alive it's about figuring
out what took
you
there
not what took the living system there so
here's the living system
but we can even finesse that it's you
and so the red line is the system
looking back how did i get here
or
putting kind of in a second person way
looking back at another lifeline how did
i get here
the question that is often asked by
living systems including us is is
something alive that's answering
schrodinger's question ramstead at all
2018. it's active topic of discussion
and so here's the line of another system
and so it's looking at the past the
present and the future of some other
system
and
is the fep
about this red arc
of the system projecting back and asking
what took
me
there
or
is it looking out to another system
and asking whether it's going to be the

ruler of like the meter stick

of

what is alive or not so probably other

ways to represent it any thoughts on

that dean yeah it's just what is alive

thing now can be can be

placed beside and when is alive and i

think that's what this paper really

brings forward

if we look only at what is alive we may

over reduce

may

but if we include when is alive as you

said things get real interesting really

really fast

and a really influential paper was

fristen's single author 2013 life as we

know it and so that's often seen to be

making a claim

about the blue because it's like real

life as we know it this is like how we

know what's alive

but then it's like even in the title

with so few words as we know it

right so even in there there's a little

bit more depth

so this is a nice road trip there's only

a few stops few sections and figures so

there's an introduction

and then minimizing free energy for

better or worse basically just like you

could minimize your potential energy

right off a cliff you can minimize free

energy right off a cliff

so

for better or for worse assuming life is

better we prefer to be alive

there's a section on the conceptual

distinctions between bayes and the free energy principle which is very didactic
there's a numerical example presented in formalism as well as in a figure which we'll talk about a lot today
and then free energy on a wing and a prior
the last section is called future direction free energy minimization as a historical scientific principle which i thought was exciting and appropriate for the paper being in biology and philosophy
so for those who are not steeped deeply in the philosophy game there's some different terminology but that's why the whole field is enriched by people bringing these different perspectives to bear
so the key words that are brought up prediction and post addition before and after historical sciences the free energy principle and then i think we added active inference and strong and weak claims they're argued they were given as a keyword but they are key words for this discussion
so let's start with prediction okay i'll just say what i have on the slide then dean our resident prediction matter expert so um first never make predictions especially about the future casey stengel or as they say in trading either predict the price or the date but never

both
and on the wikipedia page for prediction
there's so many different
related areas and that just tells you
like well what is it that's related to
all these areas things that are spooky
and forward thinking like an omen or an
oracle
like almost metaphysical or supernatural
all the way to things that are very
technical reference class forecasting
regression trends expectations
forecasting
and i think in active there's
multiple ways that prediction comes into
play so just listing only four of a
certain set
there's the prediction or the now
casting of sensations and that's like
predictive processing like the top down
priors are
modeled as providing predictions about
sensory inputs
there's the prediction
of or inference of
hidden causes in the world from those
sensory inputs which are not directly
observed
there's uh and importantly notice that
these two predictions are actually
almost instantaneous
and then we also talk about prediction
through time like at time t
there's the prediction or the
anticipation of
states at a future time and those could
be internal external sense action states
so predicted

future states

and then there's also the prediction of
the consequences of action which is
anticipatory cybernetics and can that
that can include um

counterfactuals so what if

something were to happen that's like a
prediction

about a counterfactual

so there's multiple ways that prediction
and active inference come into play

what do you think about that dean

yeah i think it's all true i think it

just points

if we're

if we're going to ask how time is

included

then

we can break out how time has been

included like what the difference is in

terms of how time is included in an

expectation versus how times you is

included in a trend estimation

because i think that's what causes us to

be able to see multiple different ways

of prediction

so not just types of prediction but then

the actual

time given over to

that particular type

great

those who study prediction

might be considered um future studiers

or futurologists

and that's also an interesting wiki page

and search area

just like how many frameworks were

brought up i mean they forgot one it

starts with active and ends with
inference
but
there's a lot of different
methods here and i'm sure we could all
think of more but it's kind of like we
have prediction as a concept looking
forward in time or even to the now which
is kind of forward in time from when we
made that prediction
and then there's the field around that
because we're going to also look at
post-diction
and then the fields that deal with
post-diction
so just whether you're hearing about
active inference for the first time or
not think about what is prediction
fundamentally looking forward in time
and then what are the attributes of
fields
that look forward in time
for example they may not have proof
because they're making a prediction
about something that hasn't yet played
out
so they might
be
an estimate that you'll bet the house on
but can one say that they have proof
when they're talking about something
going forward in time
now
now this is where it gets really
interesting because if you're an
archaeologist
slash anthropologist and you're trying
to figure out whether

the plains of salivary where
salisbury was the originating place for
all those rocks to get tipped up
you're actually looking
forward into a future
that your you think is maybe unresolved
even though if you're looking into the
deep past as well so again
i know that time time 0 seems to be
going only one direction but our minds
don't seem to necessarily honor that
exactly
so we can
contrast
prediction with post-diction
and so quoting the paper
as a reasoning pattern the fep can be
used to generate postdictive statements
in the sense of first in it all 2017
which we'll talk about accordingly fep
reasoning might be interpreted as a
principle akin to those found in the
post-dictive sciences aka historical
sciences hashtag keyword like geology
paleontology
archaeology or any science that deals
with irreproducible causes
cleland
postdictive scientific statements are
concerned with what must have been the
case instead of what will be the case
so
clelland's paper from 2002 is posted
here
and
it's very clear to see how the paper uh
under discussion draws on this one and
just to

read the conclusion of the abstract i
show that these different patterns of
reasoning
are grounded in an objective and
remarkably pervasive time asymmetry of
nature so there is this like entropies
arrow times arrow
and that makes talking about the future
and the now different than talking about
the past
that's the historical sciences
and then the first general 2017 paper
now this is interesting because
postdictive was only used one time in
the paper
figure 2 illustrates the recurrent
nature of the message passing that
mediates this predictive
and postdictive inference using little
arrows
one can clearly see that the first
outcome can influence expectation about
the final hidden state and expectations
about the final hidden state reach back
and influence expectations about the
initial step
this will be this will become an
important aspect of the deep temporal
models considered later so here's that
figure too
it's the partially observable markov
decision process that we know and love
with a layer of the message passing
added
because um
par and devry are authors and this is
like a very
i mean 2017 wasn't so long ago but this

was an important work
that combined a lot of different areas
together
and here's that reach around
through time
in an interesting way
dean
yeah this is where bringing your history
and my history into this interaction
creates a new history
and i know that's symbolically
i'm not made that's a gross over
simplification but if you were trying to
share this with
somebody with not as much background
maybe that would be a simple
simple tell i don't know
cool
and just like we talked about future
studies
let's think about the historical
sciences
so here we have the past present future
and historical sciences can refer to
history proper
the auxiliary sciences of history or any
science that draws from records of past
events
aren't experiments also past events
now the auxiliary sciences is include
but are not limited to
and there's so many fun fields here
things that are like kind of like
subfields of history like heraldry
um
paleography
so there's some great you know
vexillology

a lot of awesome terms in here
and then
i was thinking well
you know
you can't prove something in the future
and it hasn't happened yet
and then historically you also can't
prove it
because it couldn't have been otherwise
hence
you have to only say what must have been
the case instead of what will be the
case like you can't generalize you can't
say well the galapagos are biodiverse
because
and then just give one singular answer
even putting 10 virgin aside and
everything then
so what's left
the past the present and the future and
then the now is just this like kind of
moving indicator that we're experiencing
and so it doesn't really make sense to
prove something only in the now either
so that's just very interesting how
there's that
as clelland calls it the pervasive time
asymmetry of nature and you know put
your hand down if you disagree
but
what does that leave for us to actually
work with in terms of scientific
modeling
and from that tension is actually where
fep and active are going to be doing
some interesting things
yeah this was if you're this is what i
love your the

the images that you put up here because
one of the things from from a learning
standpoint that i found really
helpful
is instead of a test being what answers
can you
can you pull from your memory bank it
was a learning exhibition where the
chronology
was kind of the slow hand you'd lived
out
x number of months of
working in a professional setting and
now
in front of a significant audience you
were laying out a chronological timeline
where the focus was on
this the feelings now now top down
organized and narrated but that whole
that entire picture of feelings of
working in a professional setting for
months
actually now came out in terms of a
narrative so
the historical sciences piece here is
actually an amazing consolidator as well
and that's why that's what i found about
this paper it was actually an incredible
consolidator
agreed
so necessity and sufficiency
there are different words they mean
different things and sometimes it's easy
to
get it right in a simple case but then
when science gonna science it gets like
really mixed up
so

necessity

means that you need it like the car
needs wheels to roll normally if you
take it away it's a loss of function
the wheels are a necessity for the car
but are wheels a sufficiency for the car
well no because just four wheels doesn't
equal a car or you could have all the
car but if there's no gas like there's
something else that's also necessary
so

they're very

similar terms they both describe the
conditions

that are sufficient

and or necessary for something being the
case

and that's been described discussed in
the math threema

in this figure in the mathematical case
of course necessity insufficiency is a
key topic

in the

gene regulatory network case

that i've studied where they talk about
genes being necessary or sufficient for
a given phenotype and then also in a
legal

case from this yale law journal

talking about the most restrictive to
the least restrictive like if somebody
had no causal role

in a crime that inarguably occurred
they're blameless

if somebody was necessary and sufficient
for something to be
caused

it makes sense that they would be guilty

of that being occurred
but then there's complex systems in
between and there's the difference that
makes a difference and there's so many
ways to talk about necessity and
sufficiency
so there's sort of a simple bookmark on
both necessity and sufficiency and
neither
and then there's a complex space in
between
all i would say is no matter how often
you talk about causation it's amazing
how much stuff just gets roman candle
information wise
out of that simple word that's all i
need to say about that
the last
key word proper is free energy principle
and check out
active stream number 14 number 32 number
26 i mean
roll the die
we talk about it a lot but it's always
important to go over it because
it may be someone's first time hearing
it and the simple aspects of it are the
parts that we want to return to again
and again kind of like a strange
attractor in our model
kind of like the fep
we've talked about some of the technical
part on the bottom so check out
especially 32 and 26 for the bottom
parts
and then the free energy principle is
this imperative
to be self-evidencing and

anti-dissipative if i could only choose
two words potentially those would be it
and they're both going to come up today
because anti-dissipative will be equated
to persistent duration of survival
and then evidencing is going to be
related to
bayesian approaches to modeling evidence
and then where do you go from there
there's so many places to go active
inference the corollary process theory
and um
we're going to talk and unpack fep in a
few different domains so good to move to
the next
slide all right so
how does fep free energy principle
relate to life in this paper
so
the paper says
minimizing free energy is the process
whereby one maintains one's structural
integrity in phase of environmental
perturbations by revisiting one's most
probable organization of physiologic
states
it is in that sense that minimizing free
energy is considered a condition for
life
one can equate survival with life since
one supposes the other under the fep
and then kind of two slogans
cool sentence structure if i survived it
means that i maintained my structural
integrity in face of environmental
perturbations
in a dissipative universe according to
the era of time

and maintaining my structural integrity
is what qualifies me as living
from the point of view of the fep when
considering metamorphic organisms it may
be said that this the life cycle that
corresponds to the thing
whose integrity is maintained over time
eg over evolutionary time
not the specific form
that the system takes at one stage of
its development
e.g the adult form of a frog or ant
this casts the fep within the realm of
process ontology versus substance
ontology and i just have to say that
bucky fuller talks about structural
integrity as being like the house
that stays up through time
but
pattern integrity is in his ontology
a system that recapitulates itself and
so it retains dynamical integrity
and um then
that's connected to the idea of process
ontology versus substance ontology which
we'll discuss more on the next slide but
it's just awesome like
this is an ants an ant has a life cycle
but then you can almost pull back and
say well really the integrity of this
egg becoming an adult is related to the
nurses feeding that egg so even if that
egg to adult transition reminds us of a
solitary organism
the integrity that's being regenerated
generation after generation
is like the colony
and so

what are we talking about the
persistence of
when you never step into the same river
twice

like in one sense there is no integrity
being maintained because things are
always changing and so
the fep framing life in that way is very
interesting what do you think
yeah

this is where it's

i would often say to people i want you
to go into this situation i know it's
new for you but i want you to go into
this situation with both eyes open
and with one of the eyes i want you to
see the project at hand potentially and
with the other i want you to be able to
see the system at hand potentially
that's very confusing because
i mean the the best that we most of us
can do is just deal with the complexity
of the project or the complexity of the
system not the complexity of both
co currently but
get

having said that
the better you can
work at knowing both
are present regardless of what you
decide is your focal point in that
particular moment
the better you'll have a an opportunity
to sort of get the whole picture
going on here it's hard it in fact it
takes
it takes some moving away from the way
that most people are trained to try to

look at things so
it's really hard to be honest
let's dig into process versus substance
ontology
perfect so the 2020 dupree paper is
cited
life as process and also just a cool
kind of coincidence that it's a
bilingual
abstract if not paper with english and
russian
and
check out the paper and all the
citations if you want to read more but
just to read the highlighted parts that
kind of get the essence of the abstract
the metaphysics that has dominated
western philosophy and that currently
shapes most understanding of life and
the life sciences sees the world as
composed of things and their properties
while these things appear to undergo all
kinds of changes like metabolism
development context changes
it has often been supposed that this
amounts to no more than a change in the
spatial relations of their unchanging
parts
from antiquity however there has been a
rival to this view the process ontology
associated in antiquity with the
fragmentary surviving writings of
heraclitus
for process ontology what most
fundamentally exists is change or
process
what we are tempted to think of as
constant things are in reality merely

temporary stabilities in this constant
flux of change eddie's in the flux of
process

my main claim in this paper will be that
a metaphysics of the latter kind is the
only kind adequate to making sense of
the living world

and also related research is this
causally powerful processes from earlier
in 2021 by dupreeh in synthese
so this is a fascinating line of
research

about

moving from a substance to a
process-based view

and where causation comes into play with
that

with respect to the historical sciences
there's so many other ways to take this
another quote that one of my philosophy
professors would always say and i think
it was a quote from kant was there will
never be a newton for a blade of grass
because there isn't a mechanistic
explanation

that's timeless for a blade of grass in
the way that there plausibly could be
for like newton's cradle

so i wanted to ask you did you find this
have you known about this before reading
this paper or did the paper

um cause you to go and look and find
papers on process ontology because
two of our colleagues

talk about flux blue and and stephen
a lot and so when you brought this
forward i thought okay so has daniel
been hiding this from me all along

because he's kind of known about this
network or did the paper
in effect
create a situation where you wanted to
know more about process ontology
i followed the stigmergy
right
in the paper it says it's unclear
whether um it might be argued life and
survival refer to an enduring process
that advocates of process ontology would
call organism
in the sense of depre 2020
went down to the citations there was
yeah i knew this philosopher i knew this
philosopher but i hadn't read this paper
yeah so exactly process ontology now
generating new process ontologies
variation upon variation right and still
in the fep
still at that fep narrow time frame
it's it's that's goes back to as i said
i'm not sure if it's um just subjective
or subjective and objective
so anyway i'm glad you found this
because now i got to read dupree
because yeah
anyway oh substance where are they sting
so
how does the fpp relate to active
inference
we're kind of
getting ahead in the paper in one sense
so we'll be coming back around to it but
the fep on its own is a principle
namely a foundation for reasoning about
things eg living things
in this paper we approach the fep as

such however the fep can also be read
more broadly as a research program that
uses fep reasoning patterns to generate
scientific hypotheses
this involves implementing
fep reasoning into a theory called
active inference which is routinely used
to study various cognitive functions
when you go to the footnote
there's only one
it's important to note that fep includes
processes other than free energy
minimization
it also includes expected free energy
minimization and generalized free energy
minimization
while minimizing free energy endows the
organism with post-dictive inference
which we're gonna go into in the example
and talk about more
minimizing expected free energy endows
the organism with predictive inference
so
fep is postdictive
it looks
at the oncoming stream of data
and it compares that to basically what
must have been the case
but active or any other predictive
theory that's about expectations
it kind of has to be different because
the consequences of action are in the
future
so it has to be these two different
modes
and then
we're gonna come back to these ideas but
fep reasoning yields historical

hypotheses and not just like epic or
legendary but they're about the past
because it operates a historical
evidentiary pattern
and provides a compelling unifying
causal story
it operates a curious evidentiary
pattern though
because it assumes that both the
investigator and the thing under
investigation conform to that
evidentiary pattern
that pattern is free energy minimization
per se
right
so if you're making fewer prediction
errors
you are your your
time frame
carries on but you actually do become
[Music]
i don't know if you want the label but
you do become
somebody more versed and post-dictive
um
what do we want to call them
post-dictive
um confirmations i don't even know what
to
how to describe it
but this was a huge thing for me
post-dictator
yeah
i guess
yeah we talk a lot about prediction
markets but where's the post-diction
market
and what if

organisms

are so good at post-diction it looks

predictive

so let's see yeah it's going to be you

know they've got disclaimers whenever

you read

the forward-looking statements right

like this it

postdictive is now a new

term that i can

i can pull into those conversations

around the disclaimer it's like if

something's not predictive

it's either about right now

or it's post-active

right

let's get to the core

terminology that axel is bringing up

here and like this is what reading the

paper means is to understand like why

this entailment problem is framed

what responses exist to that problem

implicitly explicitly how it's addressed

in this paper and what the implications

are going forward that's like you'll

have read the paper to have known those

things

right

this was an earlier quote that sometimes

arguments in the literature in the fep

give the impression that in order to be

alive eg sufficiency

one must minimize free energy

such a claim does not straightforwardly

apply to the fep however and this is

what the paper will demonstrate

minimizing free energy does not entail

life

candles minimize fe too

rather the argument is that if you're
alive it probably means that you have
done something like minimizing your free
energy

which is the bay's optimal thing to do
when your life depends upon solving
complex inference problems

this is a subtle but crucial point to
getting the story straight i shall call
this the entailment problem that is the
confusion in the entailment relation
between first between free energy
minimization and life

here the notion of entailment refers to
the implication

i.e first order logical property between
free energy minimization and the fact of
displaying some life-related processes
so

let's think about two different islands
minimizing free energy

and life so we could think of it as like
a venn diagram they could be overlapping
they could be partially overlapping they
could be non-overlapping arrows can be
drawn anywhere as long as you label them
actual frames there to be two different
kinds of

threads like two archetypes of framings
in the literature

the first is the strong claim

the strong claim is that or what's
called the overly generous claim by
kirschhoff and froze 2017

is

minimizing free energy is a sufficient
condition for life so we're going to go

out there with our free energy
thermometer
and things that are minimizing free
energy are alive
because it's
sufficient
the other
[Music]
claim is the weak claim
if a system is currently alive it means
that it minimized its free energy in
other words minimizing free energy was
necessary
but not necessarily sufficient
for being alive
such a claim does not assume the fap is
designed to set the bar for the
sufficient conditions of life or meant
to predict what things may or may not be
alive
rather it limits the scope of
application of the principle to beings
that we think are alive now
what made us think that that's not in
the theory
and enables us to know the necessary
conditions under which those beings can
be living i.e can actively resist the
loss of structural integrity
what took them there
right
what do you think about that well i just
think it's interesting that he talks
later on we'll get to this is that that
you need the correct or right priors and
and if you go with a strong argument you
assume that the right priors
going forward

will continue to just
materialize versus the weak argument
which says it's not until afterwards
whether we know whether or not the right
priors were the ones picked
yeah
there's so many fun examples so let's
talk about bayesian organisms
bayesian approaches to animal behavior
whether it's bayesian approaches to
foraging or the bayesian brain
proposed that one can model
red light
one can model
not that organisms are
right one can model so just to forestall
any kind of map territory distinction
which we always return to but can't be
returned to enough
one can model
organisms as representing their relation
to environmental states using priors and
a likelihood
let's call those representations
bayesian beliefs
not in the organism it's our
representation of that in the model
on the basis of those beliefs organisms
generate adaptive behavior
bayes beliefs can represent the
probability of environmental states
prior to observing a signal that's
called a prior means before
and
the relation between environmental
states and the observed environmental
signals aka a likelihood
bayes theorem from which terms such as

the prior and likelihood come from is typically expressed as an evidentiary relationship between some prior hypotheses $p(h)$ and the observation or hand or data that's e so e is going to be like evidence or empirical h is for like hypothesis and the way that this sentence is read this expression this is bayes theorem and it's read as the probability of the hypothesis given the evidence vertical bar means given or conditioned on so the probability of the hypothesis given the evidence equals there's the equal sign both sides are the same the probability of the evidence given the hypothesis multiplied by the probability of the hypothesis that's in the brackets divided by the probability of the evidence that's bayes theorem people use bayes theorem to model all kinds of things about different systems the free energy principle is a bayesian formulation of the manner in which organisms infer the posterior probability of their prior beliefs after having a verve observed an environmental signal with a given likelihood and in doing so infer some hidden or unobserved variables same equation as the last slide what stands for the h are unobserved

variables

whose prior probability forms the hypothesis and what stands for e are the sensory signals organisms receive the data so this is like i think it's equally likely that it's day or night like it's 12 and 12. so 50 50 on whether the latent cause of the world the hypothesis is daytime or night time but you can't observe daytime or night time all you get are photons on the retina so then

it's like

you have a prior that's like 50 50 and then evidence comes in photons and then if there's a lot of photons you update your prior to get the posterior and then you think okay it's daytime because i'm getting a lot of photons hence it is often said that under the free energy principle organisms are viewed as embodying hypothesis belief or best guess about the cause of their sensations or sensory signals that they receive

that's because the evidence

for an organism are its onboard sensors and so the hypotheses are about causes of what is giving rise to

sensation just like in the daylight

example it was a hypothesis about what was giving rise to the photons

because beliefs are embodied by the organism

embedded enacted etc

and thus the organism's own states the uncertainty and the likelihood in the prior can be viewed as representing

uncertainty inherent to the biological
apparatus
instead of uncertainty in the world
as would be the case under typical
bayesian models
under the fep uncertainty should be thus
read as reporting a bayesian credence or
credibility score
over the organism's own beliefs
as it reports the probability of a state
or hypothesis relative to other possible
hypotheses what do you think dean
well i'm gonna i'm gonna what do i think
i'm gonna ask you a question back so
is this another way of saying an
account of uncertainty
right like so that that that that sounds
like wait a second how how can we have
an account of something that we
don't know but is that what he's
i know i don't want to interrupt the
math because i really like the the way
he step by step goes through this
but i when i read this again when i read
this the first time i went oh okay well
this is an account of uncertainty wait a
second how can that be so
yes
uncertainty
is about um
the hypothesis giving rise to so the way
so so the when
it takes place over a duration because
the updating of priors is happening
through time
so it's like as
my tummy is sending me signals
the body is as if

determining whether i'm hungry or not
and then
that
um uncertainty
is the probability
of a state or hypothesis relative to
other hypotheses
yeah so the traditional models are
methods that we use to try to account
for what has been learned
doesn't necessarily take this into
consideration when in fact
this is the basis
for moving forward and for learning
yes
i think there'd be some interesting
1.1.2 discussions about uncertainty
in frequentism
versus bayesian
probability
and about what it means for a
frequentist
p-value-driven understanding of
certainty or not
i fail to reject hunger at the 0.05
p-value
versus
the bayesian credible interval or the
base factor of being hungry
is this or that relative to this
conditioned on the sensory inputs
literally
hypothesis hungry or not conditioned on
the inputs yes
yeah
yeah anyway just it was again i don't
want to take this off an intention let's
get back to the math because he does a

brilliant job of clearing this up
this is such a cool discussion the
simple answer to the question of whether
priors are objective or subjective under
the fep is that
they're subjective
where's my gavel it's it's over
as we said they track the confidence
over a system's belief
under the fep
priors do not conform to irrationality
beyond the rationality of the inference
per se but nor are they rationally
unconstrained
so
this always makes me think of like sort
of the new atheist like if you were
rational you would believe like we
believe
and then that kind of absolutist
rationality
making way for a multi-perspective
rationality like i guess different
people with different
priors and sensory inputs
implement their own local rationality
and they come to different conclusions
fancy that
yeah there's a rationality beyond
bayesian rationality comes from the way
variational base is realized by the
system and we're going to talk about
that later
it basically
embeds when
into that process
because
the approximation to the posterior in

the variational bayesian inference q
is the prior used in the next cycle of
inference

the rational constraint over priors is
the fact that the approximate subjective
posterior cue

will not only be bayesian but will also
always be the best guess relative to
what the true posterior ought to be
in short under the fep
even though priors refer to
psychological states of the system or
internal states of the system
updates of the system make those priors
an approximation of what they should
have been

had the prior been updated with exact
base

thus

it might be said that priors under the
fep cut across the objective
subjective dichotomy

they are subjective

just like the gavel said above

while satisfying a rational constraint
mandated by the existence of the system
per se

yeah this is the precise approximation
argument that comes up often meaning
you

it's not a question of do we need both
we do but how do we have both
available to us well we've got to keep
both eyes open we've got to look at both
the system and the project which is
really hard because it's much easier to
just one or the other and give our
attention over to that thing

that's why post-diction is

a little bit

easier maybe hindsight isn't 20 20. as

we'll prove in this paper but it's still

better than

than

[Music]

making that making that bet with with

absolutely no priors

we talked about substance versus process

ontology

right and then we have this subjective

objective so it's almost like substance

ontology maps with objective because

it's like the object it's the substance

yeah the what

and then process ontology maps to

subjective because subjective is

relational and dynamic it entails

something different the win yeah and

then we can kind of lean more on one

side or the other

but really it's like they're nested

within each other yeah we can't let if

we let if we drop one or the other or if

you hear people dropping one or the

other you kind of have to bring them

back to

no they're both here they're both

present and they're both available

regardless of what you choose to make

your focal point that's that's when they

go well that's like just

your opinion man

right exactly

yep so let's go to the

crux of the paper which is the numerical

example

okay

so um

there's a there's some other pieces you
know we're giving a linear ordering and
kind of jumping through the paper

we're going to think about a organism

like a bacterium that's going to infer

whether a or b occurs externally

for the organism a and b are part of the
class r

and the representations by the organism
are a or b so r

can kind of be read like receptor states

a and b are inferred when receiving a

chemical signal part of the class s like

for signal

which can be alpha or beta so s is the
signal molecule

alpha or beta

we have we assume that before observing

any signal so prior the probability of a

is p and the probability of b is one

minus p so they're mutually exclusive

and they form a probability together

then the numbers

are

given

for

a and b

the prior and then the conditional

probabilities

and then

this is where life and death enters the
picture

you got to guess correctly

if it's um

a if alpha is out there and you guess a

you live

if alpha's out there and you guess wrong
you die so you want to be like on the
diagonals you want a and alpha to go
together and b and beta but you don't
want the off diagonals you'll die

okay

let's look at it graphically in figure

1.

so the right panel is a visual
representation of the organism inferring
its prior beliefs about the cause of the
observations it makes and then it
restates um several of these pieces
so we have like the um
prior and then we have the signals a and
b out there

um here's something really important for
all you active heads out there
there's no action involved in our
example

we're not talking about active inference
in more biologically realistic
descriptions of behavior like chance at
all 2020 which is where this example is
a simplified version of

we need to discuss active inference
in active inference behavior is the
result of a different inference process
that of an action policy this involves
more priors namely about the transition
between hidden states

and often about preferred sensory
outcomes so that would entail a model
where it's like okay there's a
probability of transition between alpha
and beta out there

and then there's gonna be a preference
for alpha and then i'm doing action

selection on tumble or not
based upon whether things are going well
for me or not so that's
several more pieces to the model
right but it's a toy toy example
that's going to make the point
that minimizing free energy
is not sufficient for being alive but it
is necessary
and so we're just
distilling to the simplest possible
sensory processing example
that doesn't even have action so maybe
someone could argue that well adding
action somehow
makes this example null and void or
whatever but i don't think that's the
case
and so um this does involve more priors
and action is then distinguished from
inferring hidden states it's about
inferring another hidden variable which
is policy
so the first line is the generative
model
the joint distribution which is
multiplying the prior and the likelihood
the second and the third line these two
represent the prior and the likelihood
so here's the prior it's 80 percent a
priori that it's
um that i perceive a and 20 percent a
priori i perceive b
and then there's this mapping with a and
alpha and there's like more density
there and then b and beta more density
there and the off diagonals are like
less

the fourth line is a possible bayesian
algorithm that could be used
and then the fifth line is a variational
free energy minimization which we'll
talk more about but that's kind of this
example is we're just thinking about the
sensory processing of a bacteria or cell
that is needing to correctly guess
the cause
outside
when it's only getting a receptor
on its membrane for example
what do you think
daniel do you think people will will
know
what the difference is between the
process that you
and i and steven
walk through in 33 2 and what we're
looking at right here will they will
they know just how different
those two
representations are what do you think
it's a good question we can
re-state that this is not an active
inference model right it's also not even
trying to be the most awesome bayesian
bacteria model ever right this is like
just using bayes theorem
right to show that
simply
minimizing free energy
is not gonna cut it for being alive
under this contrived setting but um when
we were in 32 or yeah 33.2
yeah and we walked through the the
matrices where there was a preference
and there was the hidden states it was

quite different

yeah so check it out do you see do you

see co-variation

happening at all

in this

co-variation um

one example of co-variation is like you

need to believe that you have accurate

sensors

if you there needs to be a covariation

between α and α

okay

if you don't have them and you don't

assume that you have them are you not

dead eventually or pretty quickly

that's sort of the um gentle

invisible hand of selections sweeping

those kinds of systems off the table

yeah yeah okay and i'm just asking

because when i saw this i the first

thing i said to myself is my goodness

this if there was a continuum between 33

2

and figure 1 of 33 or 34 0 man this

would these two things are really really

far apart

they are

um well you know

let's

let's go to 33.2

so in 33 2

this is like looking um here's what we

had done

just like last week or whatever earlier

this week

yeah um

here

the observations

are getting mapped to this prediction
about internal about hidden states
so that's actually this a matrix
um
is also
the mapping matrix
maybe it could have been a little bit
clearer because also a is like the
hidden state that's being inferred but
um this bacterium
is just reflecting
this part right here
so the prior comes in
it's like just this sort of right angled
neighborhood where you have the prior
beliefs and then we're going to have
observations coming in
and then a mapping to the hidden states
what active inference brings in is this
whole tree
where we're going to
influence through policy selection how
hidden states change through time and
that entails the preferences free energy
minimization and the affordances
so this is like just a sub component of
active
right all right
so
we're going to assume that the
computation is conformant to bayes
theorem
and that the posterior probability after
observing the signal is going to
basically update the prior
towards what is most likely given what
was observed
we're going to infer the posterior

probability of a
after observing alpha
so the prior on a was 0.8
and then the
the probability of alpha observing alpha
contingent on a was 0.7
so if it's a
in reality then 70 percent of the time
you you hear about alpha that's what is
actually on the receptor
and then the b is just one minus such
and so seven point seven point three
okay
bayes rule is applied
and you know you carry through the math
and basically
after you've observed alpha
your probability of estimating that the
external state was a
went from point eight
to point nine zero
three two
so it didn't take you to a hundred
but it bumped you up
and then
that is bayes updating
so that's a bayesian bacterium
updating its
belief about a hidden state of the world
based upon a molecule that actually only
has a 70 30 accuracy
right and that's really important to
know
like it sounds like 70 30 it sounds like
that should almost like degrade your
accuracy
but it actually even if it were 51.49 it
would still upgrade your accuracy a tiny

tiny tiny bit now if it were

1 0 0 1

well then yeah observing it if it's

perfectly associated is going to update

your belief a lot

but even something that's relatively

noisy like 70 30

can still

make you go from 80 to 90 percent sure

this is just basically taking whether or

not you've got a decontextualized

instead of information versus a

contextualized one

and what the benefits of adding the

contextualization does i think

again maybe i'm over oversimplifying but

i think that

okay so with exact bayesian inference

um

this could be computed but potentially

that's not tractable in biological

systems especially if you start to think

about how there could be like

500 metabolites

um and that's called the black box

problem or the solipsism or seclusion

problem variously in the fep literature

so how does the fep get around that

in order to bypass the problem the fep

models that inference as a proximate

base

approximate bayes bypasses the direct

evaluation of the likelihood

and

that makes it a lot more tractable

the central claim of the fep

interesting to read is that changes

leading to behavioral and

neurophysiological responses in living systems conform to a form of approximate bayesian inference known as variational base so check out active number 26 to learn more about variational bays which basically relates to we're going to control the family of equations and then we're going to do model fitting on a family of equations and that'll be easy like we're going to constrain our equation to $y \text{ equals } mx \text{ plus } b$ linear model well then we can fit it with a least squares error it's going to be a convex optimization we can do it these are different types of problems but similarly we're going to have a certain family of equations that we parameterize and that's variational base what citations did you add all i put in here was some stuff that i've been looking at like over a decade ago now and some conversations that i've had with people like morris moskovitch and reyna around the idea that neurobiologically is there some sort of an analog between walking an elephant walking across the savanna once and not really seeing the desire line versus if a trace is going if traces are going through multiple times how does that affect potential gene folding in the neurons and stuff like

that and so
it doesn't really it doesn't really i
don't
i don't think it builds on the idea
of what you're talking about in terms of
how we can
how we might want to be able to set up
software recognition software but i but
over a decade ago
um the idea that
how something is grooved in eventually
an approximation becomes something
clearer and the energy that you have to
put into that
and that's where you had the one the
above the top cat
being maybe enter too energy intensive i
just think it's really interesting
because back
over a decade ago fmris were supposed to
be the thing that was going to give us a
clearer picture
and there were people even back then
they were saying wait a second
we want to be able to talk about when a
really blurry picture becomes a more
clear picture as opposed to this is what
a bump in in in energy
tells us they kind of tried we're trying
to kill that idea more than a decade ago
cool
now
in that example everything went well for
the bacteria right it perceived a a
perceived alpha it updated a
it's all good
um
so the following numerical example will

demonstrate that minimizing free energy
is not sufficient for life
as it can lead to the exact opposite
death
this is really key
fep haters out there
note that the scope of the following
numerical example is deliberately
limited the goal is to demonstrate
minimizing free energy can lead to
maladaptive inference when performed
with the wrong priors
all things being kept fixed
if the priors are allowed to update the
inference should lead to adaptive
behavior
so it's a contrived example this isn't
how you would do fep modeling if you
wanted your system to work
adaptivity is guaranteed
to the extent which priors match
environmental constraints
more than by the machinery
employed to perform the inference so
just saying we used bays
it's like this isn't a sales pitch you
could have the wrong priors we used free
energy minimization in our software
okay well
you're going right off a cliff
so you can think
have the prior this is fine
so it's not enough to hide behind the
methodology the priors have to be
stated and they have to be adaptive
it's not enough just to say well free
energy minimization occurred
so things worked right

no it's not just like your one stop shop
it's like a linear model
but a little bit different and more
complex

so

the following example of free energy
minimization is provided with the sole
purpose of supporting our response to
the entailment problem which is to say
supporting the weak and refuting the
strong claim

the goal is to give a formal intuition
as to why minimizing free energy is not
sufficient for life

understood as the preservation of
structural integrity

by no means should the following
numerical example be viewed as an
exemplar of the manner in which free
energy minimization operates
mathematically okay so this is not an
fep model

it's just

not it's an fep like model that we're
using to make some important claims
about fep

and so here is the um in equation four
is the free energy formulation

this is the variational inference which
we're not gonna go into um right now
but um

what we'll note is just the q of r
 q of r that r is remember the receptor
state like alpha beta corresponds to the
proposal recognition or approximate
posterior density

referred to in the fpp literature

it is q of r

the recognition density remember we're
only making a sensory model
that is embodied by the organisms not to
be confused with the p of r and alpha
which would be the joint distribution or
generative model so way back in live
stream number six

we talked about how there's like the
recognition model
which is from the data to the hyper
parameters that's from the observation
of alpha or beta to a or b
and that's related to perception the
world is handing us alpha or beta
and then we're updating our causal model
of the world

but then we're not here discussing
action

so

it's kind of a limited example it's more
like a unidirectional example but adding
in the second direction doesn't get
around the fact that perceptual free
energy was minimized and that doesn't
always work so let's look at it not work
um

so we're going to

it's like and it works because it
doesn't so let's assume our organism
operates under variational bays and
still represented a with the same level
of confidence

so here's what happens this is actually
a good variational base case okay so
here

a is observed

and the fe is calculated at 0.47

and then

in six

beta is observed or b it's a question of whether b has a higher free energy contingent on observing a right so that gives a free energy of 2.297

2792 so

minimizing free energy is better

so under the variational model we still would have selected the right outcome so this is actually the variational equation working we're going to change it in a second

variational equation works

person's like yeah

[Music]

keep going minimizing free energy leads to survival only under the right conditions which is if you have the right priors so now we're going to imagine the same scenario with inverted maladaptive priors okay so you're dumped into the desert but the priors are for um

the ocean or something so here the p of b is 0.8

so now

after there's an update

the energy free energy of representing b after sensing a is 0.96

and the free energy when representing a after sensing alpha which is like the correct thing to do is higher it's over one

okay

so the minimal free energy would be representing b after sensing alpha because you had such an overwhelming

belief

in

b

that even observing a

doesn't push you all the way to

believing a

i think there's a typo in the paper

because it says represented state b with

0.63 percent confidence but it's

actually 63 percent or

0.63

right but the point is

if you started at 80 believing in b

then you got alpha

you dropped from 0.8 to 0.63 so you're

still 63 you're less

confident but still you're two-thirds

believing in b

so in other words

free energy minimization just happened

and you just picked the wrong thing

because your priors were wrong

that bacteria is gonna die

so that's the hand sweeping those

maladaptive priors off the table that's

like evolution by natural selection

hence minimizing free energy per se does

not entail life

not under the wrong prior

right not dress like that you don't

but right it's just again making a

simple

important point

minimizing free energy is not sufficient

for being a life

it is necessary for being alive

although free energy minimization is not

sufficient for life

the numerical examples suggest there may
be an entailment that goes the other way
around if you're alive it might very
well be because you did something like
minimize free energy
so that's the weak relationship
so we kind of invalidated the strong
claim
of
sufficiency but we're actually finding
support for the weak claim of necessity
this makes the fep an interesting
epistemic
principle for researchers interested in
development
of themselves of cities of organisms in
a reverse engineering fashion if we
observe a free energy minimizing
organism still living at the time that
we observe it
just like all things that we observe are
we can trust it has good enough priors
to remain alive
like i look at the ants i think
whatever
they're doing
as the world has changed they've been
doing it for 120 million years
that's what we know about that oh well
how do they know what to do when there's
a break in the trail well the ones that
didn't aren't here and the ones that are
here did it well enough and that's like
a first pass at an answer
and then also on a wing and a prior
on a wing um
means like sort of like a few different
things but it reminded me of the winged

snowflake the 2007 paper and then moth
to a flame
and so um if you're observing these
snowflakes
the ones you observe
are going to increasingly be the ones
that have wings
that can stay solid
because the ones that don't
don't exist
so that's fep on a wing and a prior
yep
one could rightfully say that minimizing
free energy when equipped with the right
prior beliefs is sufficient for life
i.e all you need to be qualified is
living
or maintaining your structural integrity
under the fep
the point here is the choice of prior
whether under bayesian or approximate
bayesian regime is a real concern since
the entailment relation between life and
fep entirely depends on the adaptivity
of those priors
interestingly some have argued the
adaptivity of priors can also be
guaranteed by a free energy minimization
operating at the population level as a
form of natural selection
the citation there was to hesp 2019
a multi-scale proposal
of the emerging complexity of life
um with hesp ramsay constant bad
kirchhoff and fristen
and here's a figure from there
i'll say though that upcoming there will
be a paper with myself

axel blue
par in fristen that will go into a lot
more detail on this
so it's absolutely a great point and i
believe that it's the case
dean what did you write here
well i was just wondering um from the
perspective of
what a prior is
like is it available
there and in terms of
something that we can leverage off of
and then of what what
then what do we do
because we have some sort of previous
marker from which to
to infer
eventually so i was wondering if that's
if that's the essence of what an
adaptive adaptive as function as an
action
um requires but
but i think i'll wait until
we bring up that paper with five authors
because
well
yeah exactly we will discuss it um
it's like
we'll have fun things to discuss i can
think of more examples but i'll save
them
so in the future direction last section
the dissolution of the entanglement
problem you know again
write a paper if you don't think that
the entitlement problem is adequately
parsed and dissolved in this paper it's
a great paper if somebody wanted to

write it

puts us in a good position to move to

another related difficulty in the

philosophical literature on the fep

which is this time of an exegetic kind

look at these synonyms

hermeneutical there's an awesome fristen

paper about the hermeneutics of

communication

right

these are just nice adjectives

if minimizing free energy is not

sufficient for life or survival

how should we interpret statements such

as

the minimization of free energy may be

necessary

if not sufficient characteristic of

evolutionary successful system systems

fristen

and steph in 2007

way way way back

[Music]

postdictive scientific statements are

concerned with what must have been the

case instead of what will be the case

like saying that

the hawks gene duplication

allowed for this body plan to change it

doesn't say about what will happen but

it is about what was necessary for

happening

so a statement such as the minimization

of free energy may be a necessary if not

sufficient characteristic of

evolutionary successful systems this

quote

is probably such a post-dictive

statement

the statement should be interpreted as
claiming that free energy minimization
must have occurred if a system is
evolutionarily successful

not the other way around

we're looking at living creatures and
postdicting

what got them there

not

looking at

the class of all systems hypothetical
failing etc and then saying what will
make them alive

because you can again minimize your free
energy right off a cliff

burn yourself out like a candle that'll
minimize free energy too you could have
maladaptive priors and also be
minimizing your free energy

daniel do you think this talks about the
difference between

deductively

realizing when and inductively

realizing when or is that

let's let's let's talk about um

deduction

induction and abduction

i think that's the win the win piece of

this again i don't want to push this

too far into the future because we're

almost at the end here but i'm just

maybe that's a point one thing i'm not
sure

yeah it's a it's a definite point one

so

the final slide

um

it's also the final lines of the paper

so

um it's a little bit of a

personal you know it's interesting to

read a paper with i it's a first off one

author paper but it's also more of like

a philosophy and dupri's papers are

written in this way as well when they're

single author it's like i argue that

this so that's kind of an interesting

style that um

stem papers usually don't have

so

um

this is axel kind of bringing it all

together as a person who used the free

energy principle in the numerical

example above

the puzzling trace for which i was

seeking a causal explanation for

was the survival of the organism that

was my observation

the causal story or hypothesis for that

observation under the conditions we

imposed to our simulated organism was

the free energy principle the inference

over which led me to write the paper you

are reading at the moment

and the live stream you're listening to

at this moment

that paper functioned as sensory

evidence for my hypothesis eg when

writing down the number and seeing they

were adding up

and that paper

live stream

is the observation that you are using to

evidence your hypothesis

concerning the claim i said at the start
of the paper namely that free energy
minimization
is not sufficient for life
so if you had wildly out of whack priors
on that claim
maybe even listening to us talk you
still believe what you believe
that's okay
it's all consistent and good
fidel to the unifying grip of hypotheses
in the historical sciences the free
energy principle is meant to account for
all of that
you me and the organism under study in a
unifying fashion
so here is
axel
looking at his figure one and then we're
reading the paper and then like active
lab and the broader ecosystem
is observing that and there's other
things observing us
so fep all i see
yeah
yeah again back to the question of
whether
it can be both objective and subjective
at the same time and when whether that
fits within the
rel relatively
small time frames of what we're looking
at when we're talking about fep and this
is even a smaller version of fpp than
maybe others looking in terms of the
unit of analysis
anyway i i just think it
the way he was able to

use math
to show
there that there is a way of arriving at
knowing what is the
um numerator and what's the denominator
here was pretty
a a was clear
b it was pretty concise given the
complexity and see it's pretty
compelling so
agree with all of that
really looking forward to the dot one
and dot two of course axel and anyone
else is welcome to join in those
discussions and then we have just so
many more papers to discuss in the
coming years
we're gonna just
do what we do in the dot one and dot two
we're gonna be in the bathtub phase be
in the thick of it be in the forest for
the dot one
and then we're gonna have our our
springboard in our jumping off in the
dot two
and
we looked forward to this dot zero but
soon
we're going to be looking back on it
what we did yeah exactly perfect perfect
setup okay thank you so much
absolutely um thanks everybody for
watching hope to see you around alright
bye